

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior software engineer with 6+ years building production systems in **Node.js / JavaScript, TypeScript** and **Python**, and a daily, hands-on user of **Claude Code** in an **AI-assisted engineering workflow**. Built and operate an open agentic pipeline on Claude Code that has run 200+ structured evaluations against an evidence-gated rubric — work that is mostly judging whether an agent's output is actually right, useful and well-reasoned, not just syntactically correct, and writing that judgment down clearly. Led the end-to-end migration of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting to the CTO; wrote SDK documentation for other engineers and trained a non-engineering Ops team on the new platform. Code on github.com/zac-09.

CORE COMPETENCIES

AI-Assisted Engineering (Claude Code)

Code Review & Engineering Judgment

Evaluation Rubrics & Written Feedback

Node.js / JavaScript / TypeScript

Python

Technical Documentation & Enablement

Microservices Architecture

Zero-Downtime Production Migrations

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO — owning the architecture and the review of every cutover.
- Implemented real-time two-way sync between MongoDB and Firestore using **Node.js** and Google Pub/Sub, so both stacks served live production traffic in parallel during the migration.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a **Python** script; moved all services off AWS, saving **\$5,000/month**.
- Shipped a 3D visualisation feature (video overlay + ffmpeg) and Express Shipping (+\$40 revenue per order), improving buyer conversion.

MTailor Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Wrote **technical documentation** for fellow engineers on the new Firebase SDKs (**Node.js** and **Python**) and trained the Ops team on the Firebase dashboard and backup procedures — explaining the *why*, not just the steps.
- Built the migration script that read from MongoDB, applied complex processing logic, and wrote to Firestore.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Extended backend APIs in **TypeScript** with Firebase Cloud Functions and Express.js on Heroku; built jobs querying **2M+ Firestore records** to keep prices current from Morningstar APIs.
- Added responsive PWA pages in React and Tailwind.

Mind2matter Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and **Node.js**; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Led a team of 4 developers** building legal-tech systems for NGOs in Uganda — set the architecture and reviewed the team's work day to day.
- Architected a microservices backend for FIDA Uganda's case-management app using **Apache Kafka, Docker and Kubernetes**; built a USSD legal-aid service talking to a Node.js backend via gRPC.
- Developed a cross-platform mobile app for LASPNET with React Native and Expo (iOS & Android) and Firebase push notifications.

PROJECTS

Career-ops — agentic evaluation pipeline on Claude Code Claude Code

Built and operate a job-evaluation pipeline that runs on Claude Code: headless-browser verification gate *before* any scoring, a six-dimension rubric where every rating must cite the source text, dedup history, parallel batch workers and Node.js PDF generation. 200+ evaluations; a large share of the work is reviewing and overriding agent output (e.g. tightening an over-loose fuzzy-match heuristic that merged distinct records).

Claude Code · Node.js · Playwright · YAML · evaluation rubric design

Healthcare WhatsApp Chatbot OpenAI + Vision

Conversational assistant reading prescription photos and responding over WhatsApp Business API — OpenAI for dialogue, Google Vision for OCR, with media handling and platform limits designed in.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore sync with idempotent, redelivery-safe writes, running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda (English-medium)

LINKS

Open-source & project code — github.com/zac-09

SKILLS

AI tooling: Claude Code (daily), agentic workflows, OpenAI API, Google Vision

Languages: JavaScript, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Apache Kafka, Google Pub/Sub

Cloud / Infra: GCP (Firestore, Cloud Functions, Pub/Sub, GCS), Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux, Git

Frontend: React, React Native, Tailwind, HTML, CSS **Spoken:** English (professional, 4+ years on US/Singapore remote teams), Kiswahili